Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 03



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Recap of Previous Lecture



Topic

Types of sets



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Topics to be Covered



Topic — Number of subsets of a set

Topic — Power set

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Topic : Types of sets

Universal Set:- It is a set of all elements associated with problem Under Consideration.

Generally denoted by U

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Topic : Types of sets

Subset & Superset:

A subset B → $A \subseteq B$

A superset B → $A \supseteq B$

Subset

Superset

- A is a subset of B if and only if B is a superset of A

Let A and B are any two sets, if every element of set A is present in set B then we say that A is a subset of B (or B is a superset A)

eg: $A = \{1, 2, a, b, c\}$
$B = \{1, a, b, 2, c\}$

Every element of set A is present in B $\therefore A \subseteq B$
Every element of set B is present in A $\therefore B \subseteq A$

if $A \subseteq B$ & $B \subseteq A$ then $A = B$

eg $A = \{1, 2, 3, 4\}$
$B = \{1, 2, 3, a\}$

$A \nsubseteq B$ & $B \nsubseteq A$

if Either $A \nsubseteq B$ or $B \nsubseteq A$ then $A \neq B$

eg: $A = \{1, 2, 3, 5\}$
$B = \{1, 2, 3, 4, 5\}$

$A \subseteq B$ but $B \nsubseteq A$

if Either $A \nsubseteq B$ or $B \nsubseteq A$ then $A \neq B$

① $\emptyset$ is a subset of Every set.

i.e wrt. any set A,

$\emptyset \subseteq A$ is always true

② Every set is a subset of it self.

i.e; wrt. any set A

$A \subseteq A$ is always true



Topic : Types of sets

Proper Subset: For a set A, any subset of set A Except set A itself will be proper subset of set A.

Set A is proper subset of set B is denoted by "$A \subset B$"

In proper subset Equality Condⁿ is not allowed

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Note:

1. if $A \subseteq B$ & $B \subseteq A$, then $A = B$
2. if $A \subseteq B$ & $B \nsubseteq A$, then $A \subset B$
   * if $A \nsubseteq B$ & $B \subseteq A$, then $B \subset A$
3. Every set is a subset of itself.
   i.e. $A \subseteq A$ is always true.
4. ϕ is a subset of every set.
   i.e. $\phi \subseteq A$ is always true w.r.t. any set A.

eg. Let $A = \{a, b, c\}$

Subsets of set A are $\Rightarrow$ $\{\}, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}$

Proper subsets of set A are = $\{\}, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}$

All subsets are Proper subsets

Except set itself.

Today's Topic



Topic : Number of subsets of a set 'A' of cardinality 'n'

Let 'A' is a finite set of size = n, {i.e. $|A| = n$}

How many subsets of set A are possible = 2^n

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$$^{n}C_{r} = \frac{n!}{r!\,(n-r)!}$$

$A = \{a, b, c\}$ $\quad |A| = 3$

Every subset of set A will include '0' or more elements out of those three elements of set A {No elements from outside set A}

Subsets of A, of size = 0

$\{\ \}$

↓

Subsets of set A of size = 0 $= {}^{3}C_{0} = 1$

Subsets of A, of size = 1

$\{a\}, \{b\}, \{c\}$

↓

Subsets of set A of size = 1 $= {}^{3}C_{1} = 3$

Subsets of A, of size = 2

$\{a,b\}, \{a,c\}, \{b,c\}$

↓

Subsets of set A of size = 2 $= {}^{3}C_{2} = 3$

Subsets of A of size = 3

$\{a, b, c\}$

↓

Subsets of set A of size = 3 $= {}^{3}C_{3} = 1$

Total no. of Subsets of set A $= {}^{3}C_{0} + {}^{3}C_{1} + {}^{3}C_{2} + {}^{3}C_{3}$

$\therefore 1 + 3 + 3 + 1 = 8 = 2^{3}$

Let $|A| = n$, then { Every subset of set A will include 0 or more elements out of those n elements }

Total no. of subsets of set A = Number of Subsets of size = 0 + Number of Subsets of size = 1 + ... + Number of Subsets of size = k ($k < n$) + - - + Number of Subsets of size = n

$$= {}^{n}C_0 + {}^{n}C_1 + \quad + {}^{n}C_k + - - - - + {}^{n}C_n$$

Note:

$$ {}^{n}C_0 + {}^{n}C_1 + {}^{n}C_2 + \cdots\cdots + {}^{n}C_n = 2^n $$

$$ (1+x)^n = {}^{n}C_0\, x^0 + {}^{n}C_1 \cdot x^1 + {}^{n}C_2 \cdot x^2 + \cdots + {}^{n}C_n \cdot x^n $$

Put. $x = 1$

$$ (1+1)^n = {}^{n}C_0 (1)^0 + {}^{n}C_1 \cdot (1)^1 + {}^{n}C_2 \cdot (1)^2 + \cdots + {}^{n}C_n (1)^n $$

$$ (2)^n = {}^{n}C_0 + {}^{n}C_1 + {}^{n}C_2 + \cdots + {}^{n}C_n $$

Let $|A| = n$, then { Every subset of set A will include 0 or more elements out of those n elements }

Total no. of subsets of set A = Number of Subsets of size = 0 + Number of Subsets of size = 1 + ... + Number of Subsets of size = k (k < n) + - - + Number of Subsets of size = n

$$= {}^{n}C_0 + {}^{n}C_1 + + {}^{n}C_k + - - - - + {}^{n}C_n = 2^n$$

No. of subsets of size at most 'k'

No. of subsets of size Exactly = k

No. of subsets of size at least 'k'



Topic : Number of subsets of a set 'A' of cardinality 'n'

Let $|A| = n$,

① Total no. of subsets of set A $= {}^nC_0 + {}^nC_1 + \cdots + {}^nC_n = 2^n$

② Number of subsets of set A, such that size of each subset is exactly 'k' $= {}^nC_k$ {where $k \leq n$}

③ Number of subsets of set A, such that size of each subset is at most 'k' $= {}^nC_0 + {}^nC_1 + \cdots + {}^nC_k$

④ Number of subsets of set A, such that size of each subset is at least 'k' $= {}^nC_k + {}^nC_{k+1} + \cdots + {}^nC_n$

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Number of Subsets

Let $A = \{1, 2, 3, 4, \ldots\ldots, n\}$

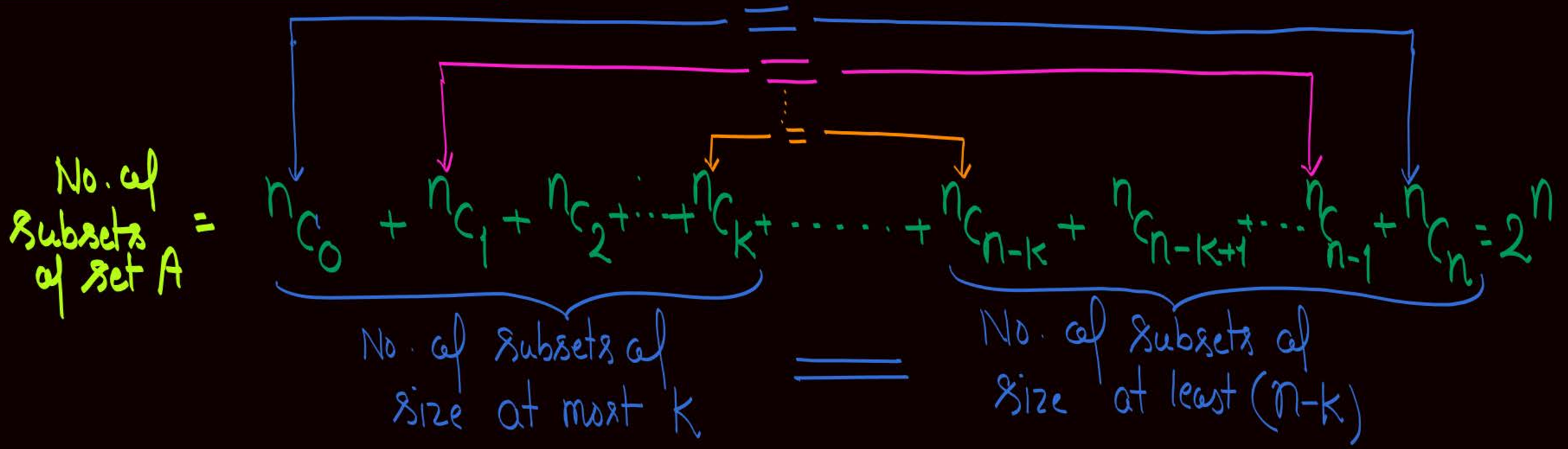
in a subset of set A, for each element of set A we have two choices
(i) include that element into subset
or (ii) Don't include that element into subset

Number of Subsets of Set A $= 2 * 2 * 2 * 2 * \ldots\ldots * 2 = 2^n$

and

n times 2

We know, $^nC_r = {}^nC_{n-r}$



Q: Let $|A| = 5$,
How many subsets of set A are there of size at most '3'.

Solun. $= {}^5C_0 + {}^5C_1 + {}^5C_2 + {}^5C_3$

$= 1 + 5 + 10 + 10$

$= \boxed{26}$

$K = 3 \quad (n-k) \Rightarrow (5-3) = 2$

No. of subsets of size at least $(5-3) = 2$

$= {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5$

$= 10 + 10 + 5 + 1$

$= \boxed{26}$

Q:. Let $A = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

How many subsets of set A are there that contains exactly three odd numbers.

- Total no. of odd nos = 5 $\Rightarrow$ We need to choose exactly '3' out of those '5' it can be done in $^5C_3 = 10$ ways
- There may any no. of Even nos in the subset $\Rightarrow$ Total no. of even no. = 6

Total no. of Possible subsets w.r.t. given Constraint

$$= {}^5C_3 * ({}^6C_0 + {}^6C_1 + {}^6C_2 + {}^6C_3 + {}^6C_4 + {}^6C_5 + {}^6C_6)$$

$$= 10 * 2^6 = 10 * 64 = 640$$

Note :.

Let $|A| = n$.

then No. of proper subsets of set A $= \underbrace{2^n}_{\text{all subsets}} - \underbrace{1}_{\text{set A itself.}}$

↓ Except

* Concept w.r.t. number of Supersets :-

* Number of supersets of a given set will always be defined w.r.t. Universal set.

Q. Let $A = \{1, 2, 4, 5\}$ is a set derived w.r.t universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
How many supersets of set A are there ?

$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

Number of Super-sets of Set A $= 1 * 1 * 2 * 1 * 1 * 2 * 2 * 2 * 2 * 2 = 2^6 = 64$

For elements of set A there will be only one choice. i.e. they all must be present

⊛ Let 'A' is a set derived from Universal set U, such that $|A| = x$ & $|U| = n$

then No. of supersets of set A $= 2^{n-x}$

$= 2^{|U| - |A|}$



Topic : Power Set

Power Set: Let 'A' is any finite set, then power set of set 'A' is a set of all subsets of set A.

Power set of set A is denoted by $P(A)$ or 2^A

$P(A)$ = Set of all subsets of set A

$P(A) = \{$ ... $\}$

it is a set

Elements will be all subsets of set A

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Q: Let $A = \{a, b, c\}$

then

$P(A) = ?$

Soln.

$$P(A) = \{\{\}, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}\}$$

Note:- ① $P(A)$ is also a set, and we know that $\emptyset$ is a subset of every set

$\therefore \emptyset \subseteq P(A)$ is true for any set A.

② $\emptyset$ is always an element of $P(A)$, for any set A.

i.e. $\emptyset \in P(A)$ is true for any set A.



Topic : Cardinality of Power Set

Cardinality of $P(A)$ = $|P(A)|$ = No. of elements in $P(A)$

= Total no. of subsets of set A.

$= 2^{|A|}$

$$|P(A)| = 2^{|A|}$$

Slide

Q:. Let $\emptyset$ be an Empty set. then

$$|P(P(\emptyset))| = ?$$

(a) 0

(b) 1

(c) 2

(d) Can not be defined.

We know, $|\emptyset| = 0$

$$|P(A)| = 2^{|A|} \quad \therefore |P(\emptyset)| = 2^{|\emptyset|} = 2^{0} = 2 = 1$$

Hence,

$$|P(P(\emptyset))| = 2^{|P(\emptyset)|} = 2^{1} = 2$$

Q: Let $\emptyset$ be an Empty set. then

$$|P(P(\emptyset))| = ?$$

(a) 0
(b) 1
(c) 2
(d) Can not be defined.

We must understand

The only subset of $\emptyset$ is $\emptyset$ itself.

$P(\emptyset)$ = Set of all subsets of $\emptyset$

$= \{ \emptyset \}$ ($\emptyset$: subset of $\emptyset$)

$P(P(\emptyset))$ = Set of all subsets of $P(\emptyset)$, i.e. Set of all subsets of $\{\emptyset\}$

$= \{ \{\}, \{\emptyset\} \}$

There are the only two subsets of $\{\emptyset\}$

$$|\{\emptyset, \{\emptyset\}\}| = 2$$

$$\emptyset \neq \{\emptyset\}$$

$\{\}$: it is an Empty set

$\{\{\}\}$: it is a set Containing one element. And that element is an Empty set

$$|\{\}| = 0$$

$$|\{\{\}\}| = 1 = |\{\emptyset\}|$$

Q: Let $A = \{1, 2, 3, \{1,2\}, \{\{2,3\}\}\}$

Which of the following is/are true

{Exactly How many of the given options are True = 11 = Ans}

✓ (a) $1 \in A$

✗ (b) $\{1\} \in A$

✗ (c) $1 \subseteq A$

✓ (d) $\{1\} \subseteq A$

it is a member of A
∴ True

✓ (e) $\{1,2\} \in A$

✓ (f) $\{1,2\} \subseteq A$

all are elements of set A ∴ True

✗ (g) $\{\{1,2\}\} \in A$

✓ (h) $\{\{1,2\}\} \subseteq A$

✗ (i) $\{2,3\} \in A$

✓ (j) $\{2,3\} \subseteq A$

✓ (k) $\{\{2,3\}\} \in A$

✗ (l) $\{\{2,3\}\} \subseteq A$

✓ (m) $\{\{\{2,3\}\}\} \subseteq A$

✗ (n) $\emptyset \in A$

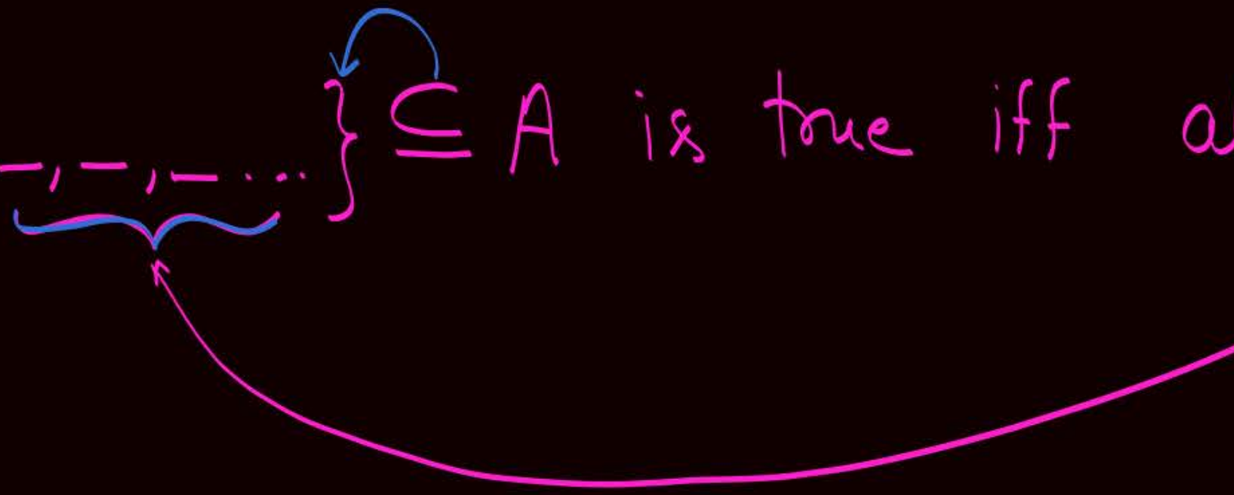
✓ (y) $\emptyset \in P(A)$

✓ (x) $\emptyset \subseteq A$

✓ (z) $\emptyset \subseteq P(A)$

- $a \in A$ is true iff 'a' is a member of set A

- $\{_, _, _ \ldots\} \subseteq A$ is true iff all elements are member of set A.



H.W.

Q1. For a set A, the power set of A is denoted by 2^A. If $A = \{5, \{6\}, \{7\}\}$ which of the following options are true?

A. $\emptyset \in 2^A$

B. $\emptyset \subseteq 2^A$

C. $\{5, \{6\}\} \in 2^A$

D. $\{5, \{6\}\} \subseteq 2^A$

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H.W.

Q2. Let $P(S)$ denote the power set of a set S. Which of the following is always true?

A. $P(P(S)) = P(S)$

B. $P(S) \cap P(P(S)) = \{\emptyset\}$

C. $P(S) \cap S = P(S)$

D. $S \notin P(S)$

E. None of the above

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2 mins Summary

Topic Number of subsets of a set

Topic Power set

Slide

THANK - YOU